MovieLens Hybrid Movie Recommender

Technical report for Project 1: a three-algorithm comparison and interactive system

Dataset	MovieLens 1M
Evaluation	Chronological 80/10/10, shared 100-candidate ranking
Algorithms	ItemKNN, LightFM-style BPR, ICSRec-inspired sequence model
Date	13 November 2026

Abstract

This project builds and evaluates a movie recommendation system on MovieLens 1M. The system uses item-based collaborative filtering, a LightFM-style hybrid metadata factorizer, and an ICSRec-inspired latent-intent sequence recommender. All models share a per-user chronological split and identical sampled candidates. The strongest held-out model by NDCG@10 is **ItemKNN** at **0.3696**, with HR@10 of **0.6415**.

A Streamlit interface accepts an anonymized user, excludes observed movies, and shows recommended titles and genres. The repository includes data inspection, trained model export, tests, figures, and full reproduction instructions.

1. Task and success criteria

Given a user, recommend at least five movies not present in that user's observed history. The implementation also must cover dataset evidence, preprocessing, exactly three algorithms, comparative evaluation, and an interactive interface.

- Reproducible temporal split and fixed random seed.
- No held-out ratings in training features or learned popularity counts.
- Identical candidates and metrics across all three algorithms.
- User-visible titles, genres, and an exploratory application.

2. Dataset evidence

The official MovieLens 1M archive contains users, movies, and ratings tables. The tables have no missing cells. Release year is parsed from titles, genres are multi-hot encoded, age is standardized, and gender and occupation are one-hot encoded. All later data-derived statistics use training rows only.

Table	Rows	Columns	Null cells	Null columns
users	6,040	5	0	0
movies	3,883	3	0	0
ratings	1,000,209	4	0	0

Ratings span 1 to 5 with mean 3.582. The source contains 0 duplicate user-movie pairs. The script exports shapes, total nulls, null columns, unique counts, dtypes, and a table comparison to the processed data directory.

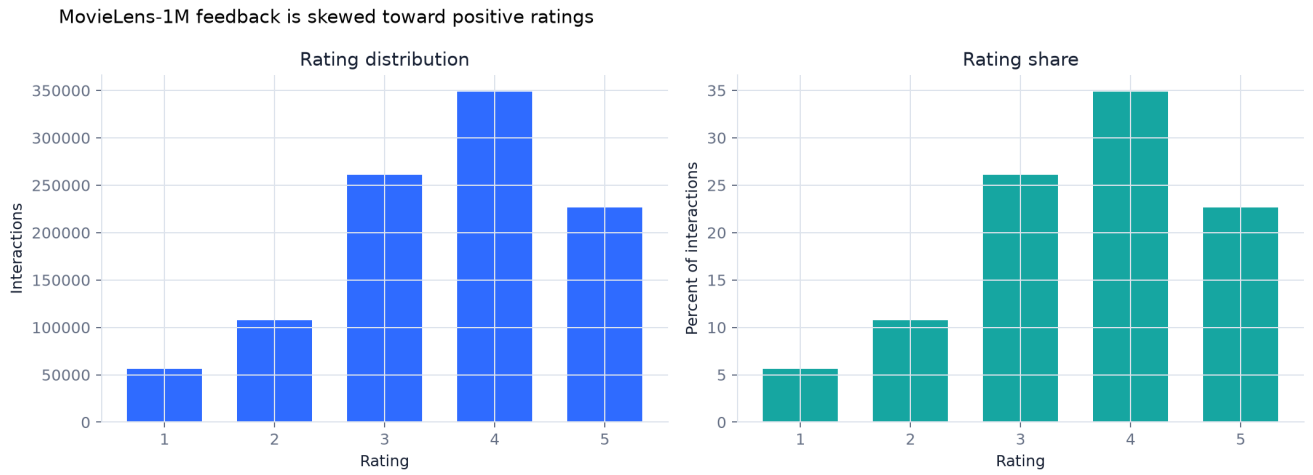


Figure 1. Ratings are concentrated in the upper half of the scale.

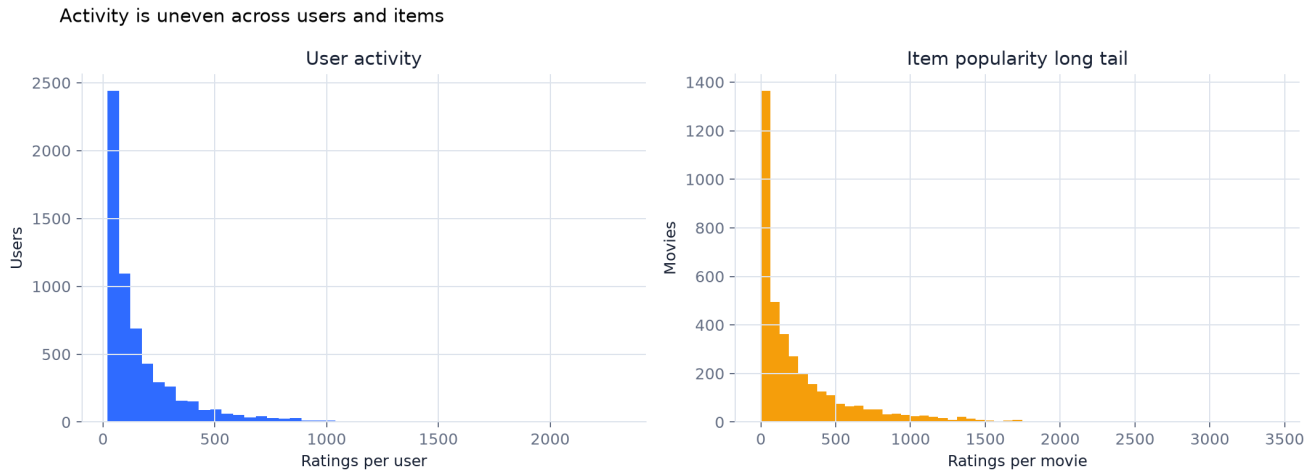


Figure 2. User activity and item popularity both have long right tails.

3. Preprocessing and split

The split contains 802,553 training, 100,273 validation, and 97,383 test ratings. Interactions are sorted by timestamp inside each user and divided 80/10/10. Item popularity, item-item similarity, sequence histories, and feature normalization are fitted from training rows only.

Ratings 4 and 5 become positive relevance labels. For ranking evaluation, each user contributes the latest relevant validation or test item and 99 movies never rated by that user. The shared seed makes candidate order identical for every model. The model receives no future rating when fitting, and the final test metrics are not used for hyperparameter choice.

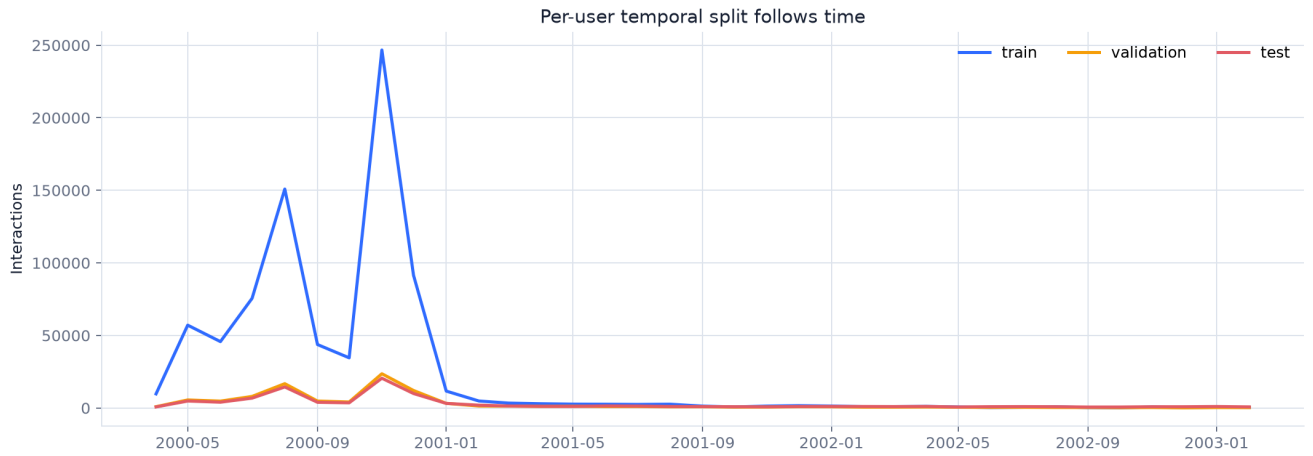


Figure 3. The per-user split follows interaction time.

4. Algorithms

4.1 Item-based collaborative filtering

A binary training matrix uses ratings of 4 or higher. L2-normalized movie columns produce cosine item-item similarity. A user's unseen-item score sums similarities from movies the user liked and adds a small normalized log-popularity prior. Neighbor counts 30, 60, and 120 are compared on validation NDCG@10.

4.2 LightFM-style metadata embeddings

The LightFM paper uses additive identity and metadata embeddings. This project implements score(u,i) = b_u + b_i + <e_u + F_u w_u, e_i + F_i w_i>. User features are age, gender, and occupation. Item features are genre and release year. Bayesian Personalized Ranking trains each observed positive above a sampled unobserved negative. Dimensions 32 and 64 are compared.

4.3 ICSRec-inspired sequential model

A GRU encodes up to 50 positive training items. KMeans refreshes 8 or 16 latent intent clusters from sequence representations. An InfoNCE loss aligns sequences with intent prototypes, and a BPR next-item loss trains ranking. This is a compact adaptation of ICSRec, not an exact reproduction of every augmentation and training detail in the paper.

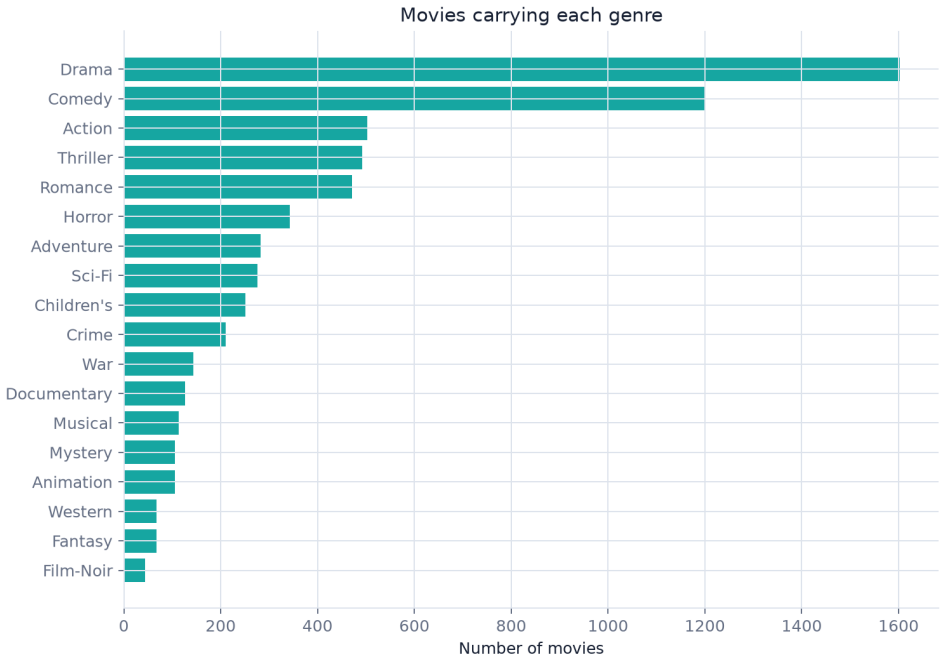


Figure 4. Drama and comedy dominate the catalog, increasing popularity bias risk.

5. Evaluation protocol

All models score the same 100-item candidate set per user. Metrics are HR@10, NDCG@10, and MRR. With one relevant item per user, Precision@10 and Recall@10 equal HR@10. Validation NDCG@10 selects one configuration per algorithm, then the selected configuration receives a single final test evaluation.

6. Results

Algorithm	HR@10	NDCG@10	MRR	Users
ItemKNN	0.6415	0.3696	0.2863	5730
LightFM-adapted	0.5640	0.3117	0.2348	5730
ICSRec-inspired	0.4806	0.2562	0.1883	5730

The best held-out NDCG@10 is 0.3696. This is a controlled 100-candidate comparison and should not be interpreted as full-catalog accuracy. The test table is written to results/final_test_metrics.csv.

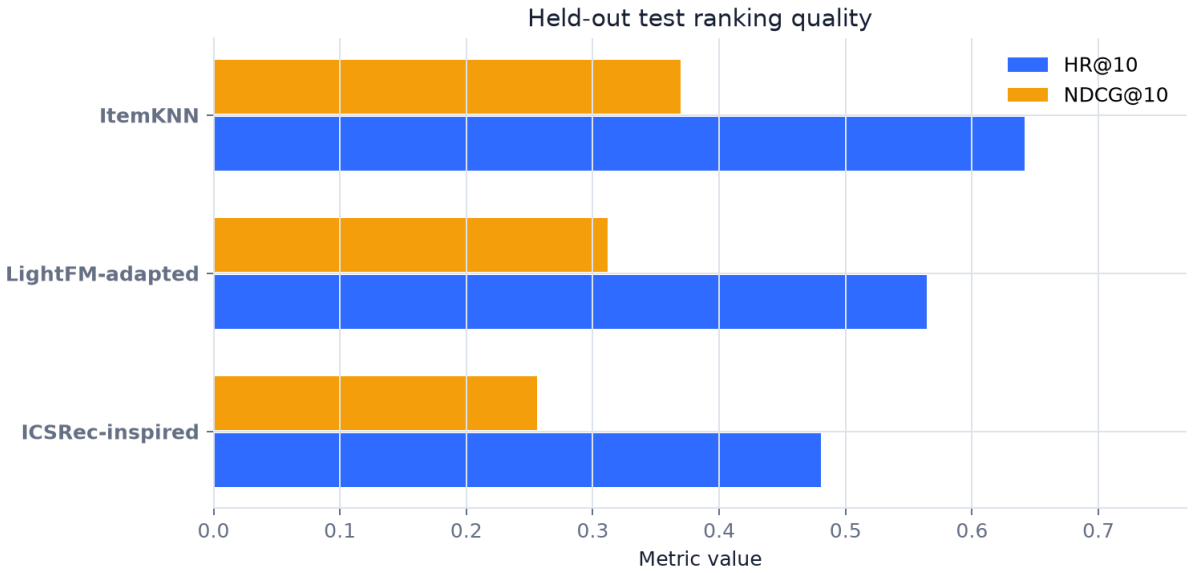


Figure 5. Held-out HR@10 and NDCG@10 for the selected models.

Validation selection

Algorithm	HR@10	NDCG@10	MRR	Configuration
ItemKNN	0.6999	0.4204	0.3344	neighbors: 30, popularity_weight: 0.02
ItemKNN	0.7132	0.4300	0.3428	neighbors: 60, popularity_weight: 0.02
ItemKNN	0.7106	0.4314	0.3455	neighbors: 120, popularity_weight: 0.02
LightFM-adapted	0.6186	0.3520	0.2706	batch_size: 4096, embedding_dim: 32, epochs: 10,
LightFM-adapted	0.6113	0.3449	0.2634	batch_size: 4096, embedding_dim: 64, epochs: 10,
ICSRec-inspired	0.5079	0.2730	0.2018	batch_size: 256, embedding_dim: 32, epochs: 6, l
ICSRec-inspired	0.5059	0.2745	0.2042	batch_size: 256, embedding_dim: 64, epochs: 6, l

7. Interpretation and error analysis

ItemKNN is transparent and captures co-consumption, but mainstream titles gather more co-ratings and can receive overly broad neighborhoods. Metadata embeddings generalize better for sparse items and users, while the sequence model can adapt to order and changing preferences. The sequence model is limited by a short CPU training budget and a 50-item history.

- **Popularity bias:** frequent items have more training evidence and stronger prior support.
- **Sparse users:** a small positive history weakens both collaborative and sequence estimates.
- **Cold items:** item-item neighborhoods are unstable, although metadata still supplies signal.
- **Taste drift:** a static embedding may blur preferences from different time periods.
- **Implicit labels:** ratings below 4 are excluded from fitting, losing negative preference detail.

8. Recommendation examples

The table below shows representative outputs from the exported top-10 sample. The full file contains 20 users and all three algorithms.

User	Model	#	Movie	Genres	Year
521	ItemKNN	1	Deliverance (1972)	Adventure|Thriller	1972
521	ItemKNN	2	Taxi Driver (1976)	Drama|Thriller	1976
521	ItemKNN	3	Boogie Nights (1997)	Drama	1997
521	ItemKNN	4	Where Eagles Dare (1969)	Action|Adventure|War	1969
521	ItemKNN	5	Jerry Maguire (1996)	Drama|Romance	1996
521	ItemKNN	6	Breakfast Club, The (1985)	Comedy|Drama	1985
521	ItemKNN	7	Get Shorty (1995)	Action|Comedy|Drama	1995
521	ItemKNN	8	Atlantic City (1980)	Crime|Drama|Romance	1980
521	ItemKNN	9	Westworld (1973)	Action|Sci-Fi|Thriller|Wes	1973
521	ItemKNN	10	What's Eating Gilbert Grape (1993)	Drama	1993
521	LightFM-adapted	1	Atlantic City (1980)	Crime|Drama|Romance	1980
521	LightFM-adapted	2	Dangerous Liaisons (1988)	Drama|Romance	1988
521	LightFM-adapted	3	Taxi Driver (1976)	Drama|Thriller	1976
521	LightFM-adapted	4	Breaking Away (1979)	Drama	1979
521	LightFM-adapted	5	Westworld (1973)	Action|Sci-Fi|Thriller|Wes	1973
521	LightFM-adapted	6	Boogie Nights (1997)	Drama	1997

9. Application

The Streamlit application accepts a user, algorithm, genre, minimum release year, and result count. It excludes every observed title, displays movie metadata, compares all three algorithms for one user, exposes model metrics, and downloads recommendations as CSV. Start it with `streamlit run app.py`.

10. Limitations and future work

Sampled evaluation does not measure full-catalog recall or business value. A production study should assess catalog coverage, novelty, fairness across user groups, and temporal decay. Text reviews and richer metadata could improve cold start. The two recent/reference methods are transparent adaptations, so parameter results are project-specific. Future work should include graded feedback, popularity calibration, cold-start experiments, and online A/B testing.

11. Reproducibility

Random seed is 42 and the positive threshold is 4. The complete selected configuration is {"ICSRec-inspired": {"batch_size": 256, "embedding_dim": 64, "epochs": 6, "learning_rate": 0.003, "loss": "BPR + intent InfoNCE", "n_intents": 16}, "ItemKNN": {"neighbors": 120, "popularity_weight": 0.02}, "LightFM-adapted": {"batch_size": 4096, "embedding_dim": 32, "epochs": 10, "learning_rate": 0.01, "loss": "BPR"}}. The repository README lists software requirements, run commands, dataset citation, and all result locations. Raw and processed MovieLens files are excluded from Git.

References

F. Maxwell Harper and Joseph A. Konstan. 2015. The MovieLens Datasets: History and Context. ACM TiiS 5(4), Article 19. doi:10.1145/2827872.

Maciej Kula. 2015. Metadata Embeddings for User and Item Cold-start Recommendations (LightFM). arXiv:1507.08439.

Xiuyuan Qin et al. 2024. Intent Contrastive Learning with Cross Subsequences for Sequential Recommendation (ICSRec). WSDM 2024. doi:10.1145/3616855.3635773.